

Brazilian Marketplace Analysis

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1. Introduction

In this assignment, we're working with a Brazilian e-commerce public dataset containing orders made at Olist Store. The dataset includes information on 100,000 orders from 2016 to 2018 across multiple marketplaces in Brazil. It provides a multidimensional view of customer orders, covering aspects such as order status, pricing, payment details, freight performance, customer location, product attributes, and customer reviews.

2. Data

2.1 Data Loading and Preview

The dataset is composed of 9 separate .csv files. Table 2.1.1 outlines each file's name, number of columns, and number of rows.

Name	Columns	Rows
Customers	99441	5
Order Items	112650	7
Payments	103886	5
Reviews	99224	7
Orders	99441	8
Products	32951	9
Product Translation	71	2
Geographical Location	821361	5
Sellers	3095	4

Table 2.1.1

2.2 Data Merge

It appears that `order_id` is the primary key across the different files. Since our goal is to have one row per customer, we first merge the orders table with the customers table to bring in `customer_unique_id`. Then we join that with `order_items`, followed by merging `product_category_name` into the items. Finally, we merge the reviews data with customer information.

2.3 Aggregation

To prepare features for clustering, we aggregate the data at the customer level. We compute the number of orders (`n_orders`), average order price, average freight value, number of unique product categories, the most frequent category per customer, average items per order, average review score, number of reviewed orders, and average delivery and shipping delays.

2.4 Data Cleaning and Preprocessing

2.4.1 Translation

Since some category names were in Portuguese, we translated them into English and normalized the text format (including case and characters). In retrospect, this step turned out to be unnecessary as the feature was not ultimately used.

2.4.2 Feature Selection and Null Handling

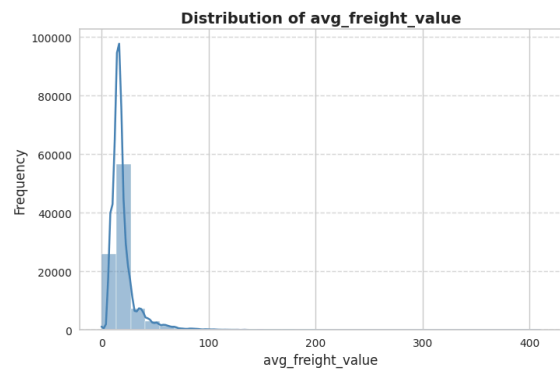
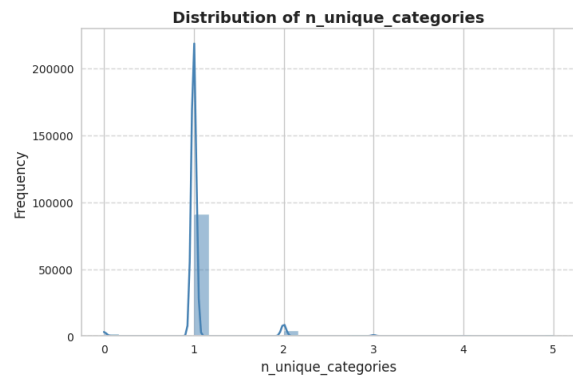
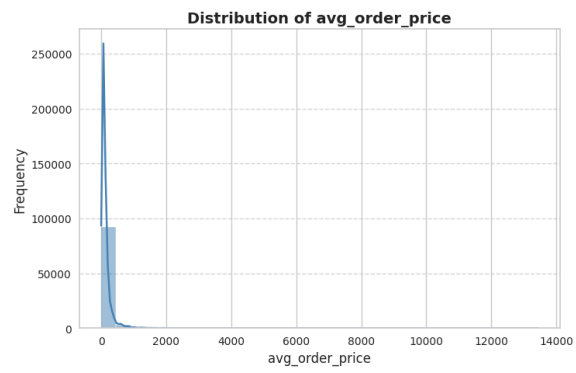
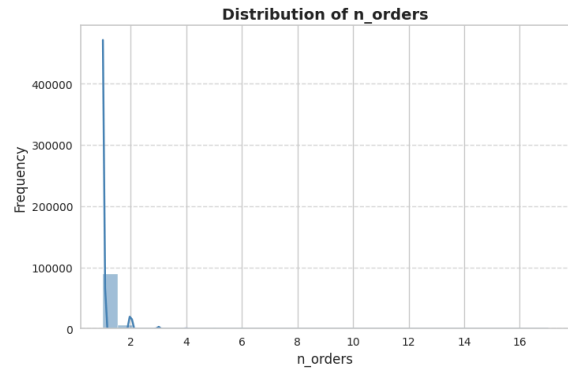
We selected features including `n_orders`, `avg_order_price`, `avg_freight_value`, `n_unique_categories`, `top_category`, `avg_items_per_order`, `avg_review_score`, `n_reviewed_orders`, `avg_delivery_delay`, and `avg_shipping_delay`. We also included `customer_city` and `customer_state`, and added a `total_spent` feature (calculated as `avg_order_price * n_orders`).

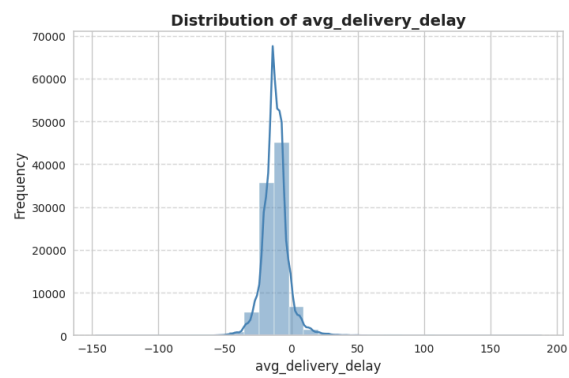
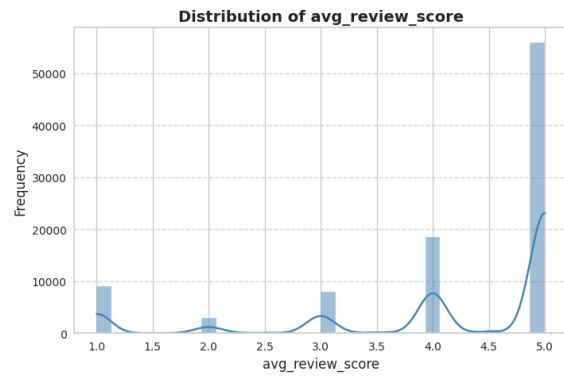
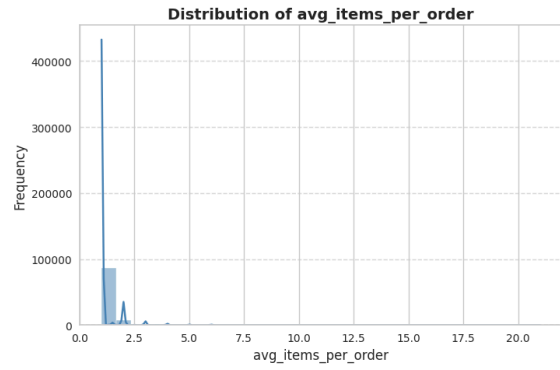
Since null values were minimal, we dropped most of them. For missing `top_category` values, we filled them with 'unknown'. The final set of columns in our cleaned dataframe included: `customer_unique_id`, `n_orders`, `avg_order_price`, `avg_freight_value`, `n_unique_categories`, `top_category`, `avg_items_per_order`, `avg_review_score`, `n_reviewed_orders`, `avg_delivery_delay`, `avg_shipping_delay`, `customer_city`, `customer_state`, and `total_spent`.

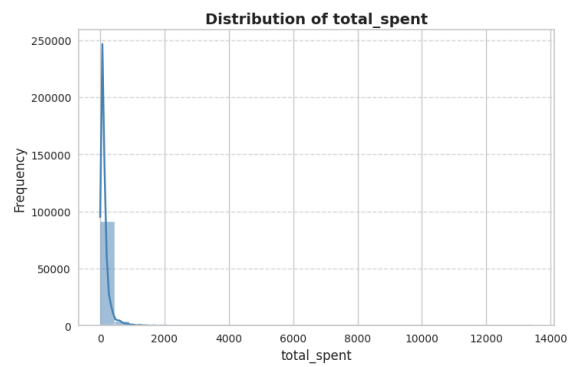
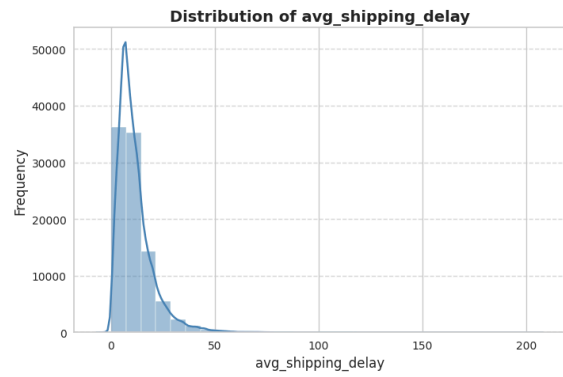
3. EDA

3.1 Distributions

We explored the distribution of our numerical features. The number of orders is heavily skewed towards 1, meaning most customers placed only one order. The distribution of average order price is right-skewed, peaking around R\$100. Similarly, average freight value is right-skewed and peaks around R\$25. The number of unique categories and average items per order also spike at 1. Most customers gave 5-star reviews, with 2-star ratings being the least common. As expected, the number of reviewed orders typically equals 1. Delivery delays appear normally distributed, while shipping delays range widely (0–50 days) with a right-skew. `Total_spent` mirrors the average order price due to the dominance of single-order customers.

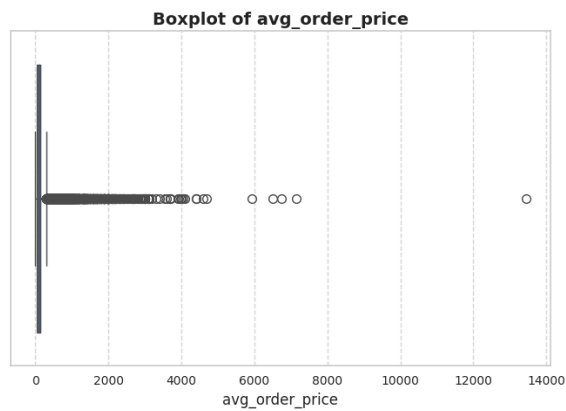


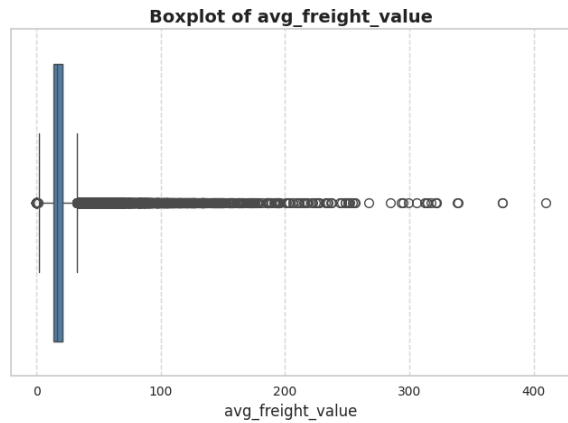




3.2 Outliers

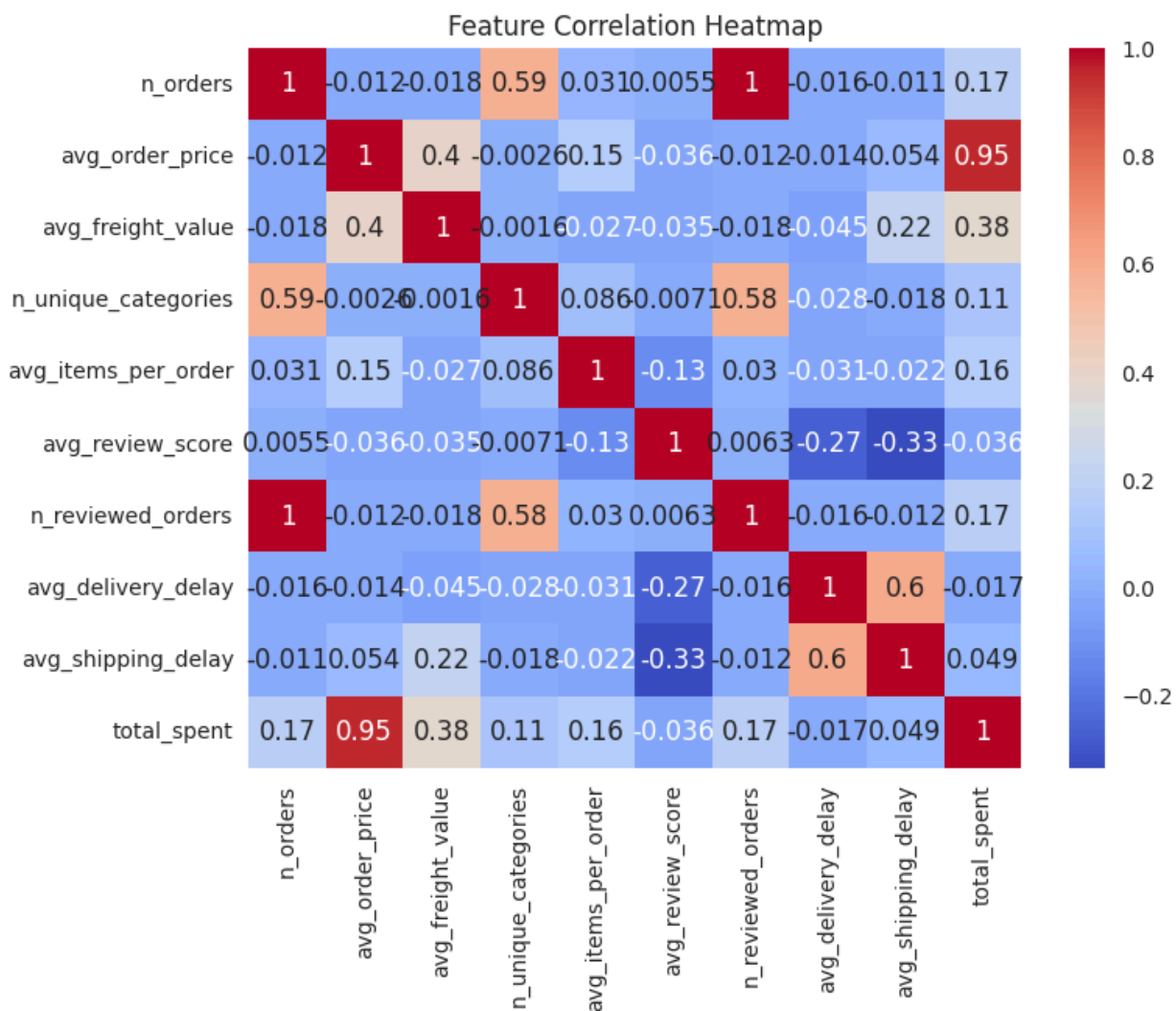
To identify outliers, we created boxplots for numerical features. These plots revealed long tails in several features, suggesting the presence of significant outliers or unusual behavior



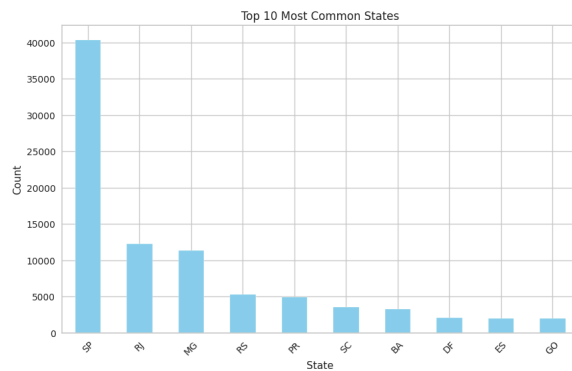
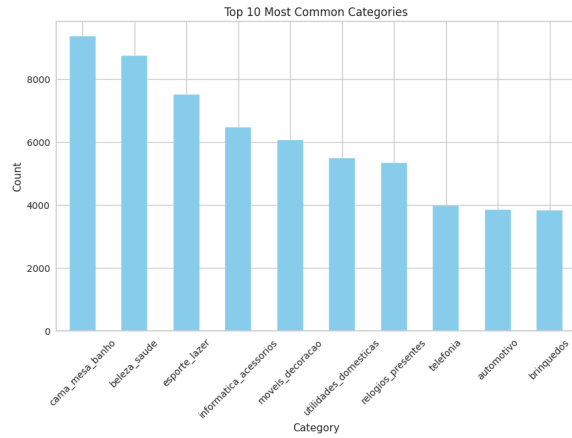


3.3 Correlation

We generated a correlation heatmap to assess relationships between features. Highly correlated features were candidates for removal to avoid redundancy. For instance, `avg_order_price` and `total_spent` had a correlation of 0.95, so we retained only one. Similarly, `n_reviewed_orders` and `n_orders` were highly correlated, as were `avg_shipping_delay` and `avg_delivery_delay`.



A frequency plot showed that "bed_bath_table" was the most common product category, followed by "health_beauty." Geographically, most orders originated from São Paulo, with Rio de Janeiro as a distant second.

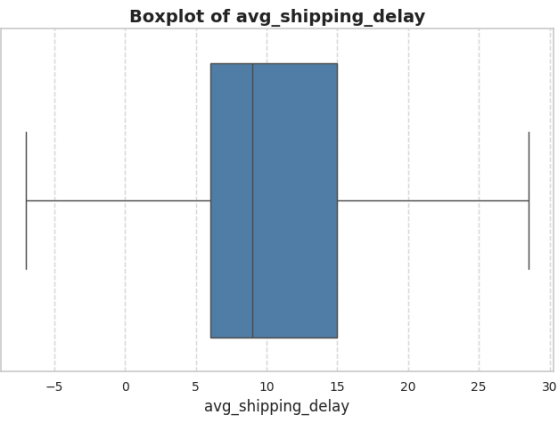
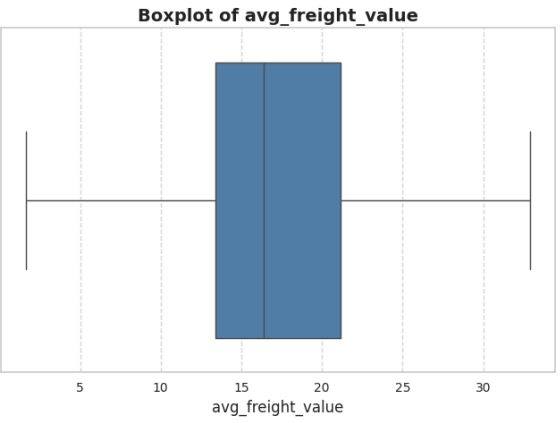
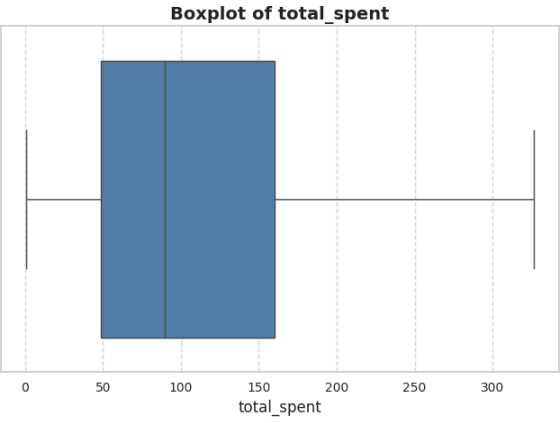


4. Handling Outliers and Feature Engineering

To manage the high cardinality in product categories, we retained only those appearing more than 100 times and grouped the rest as “others.” We also created a “VIP” customer label for those above the 75th percentile in total spending.

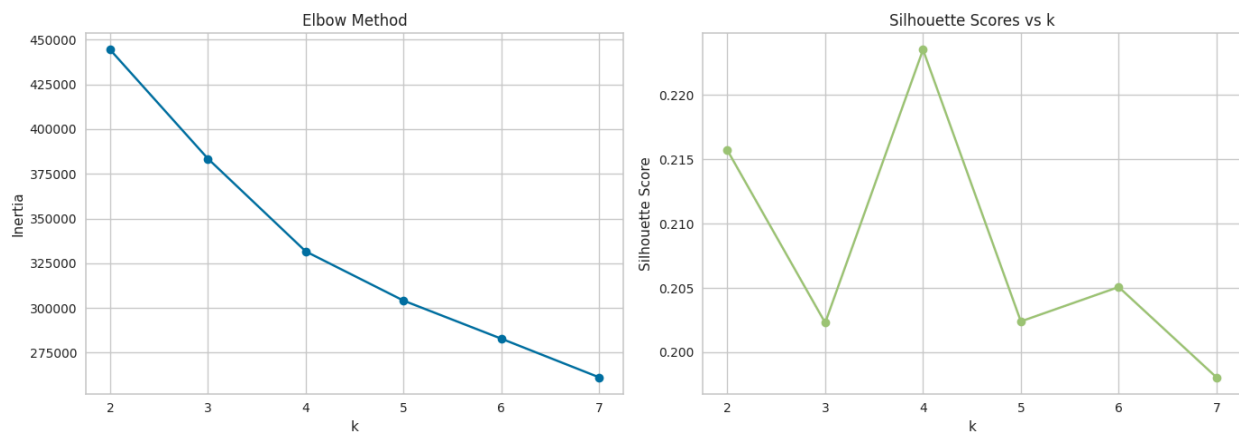
To encode categorical variables like `customer_state` and `top_category`, we used frequency encoding. This way, a customer’s state or category was represented by how frequently it appeared in the dataset.

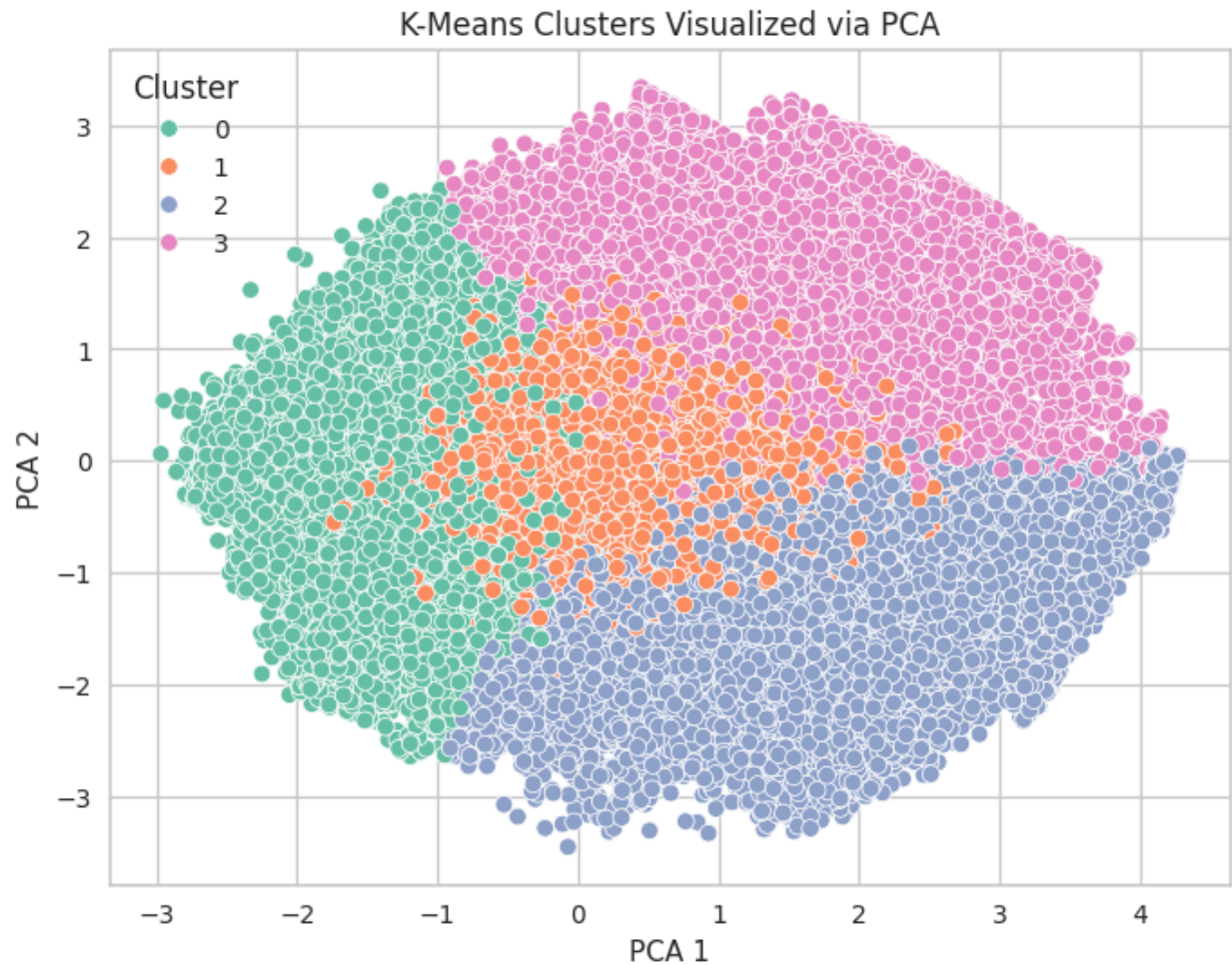
To handle extreme outliers, we capped values using the IQR method. This significantly improved the shape and interpretability of our boxplots.



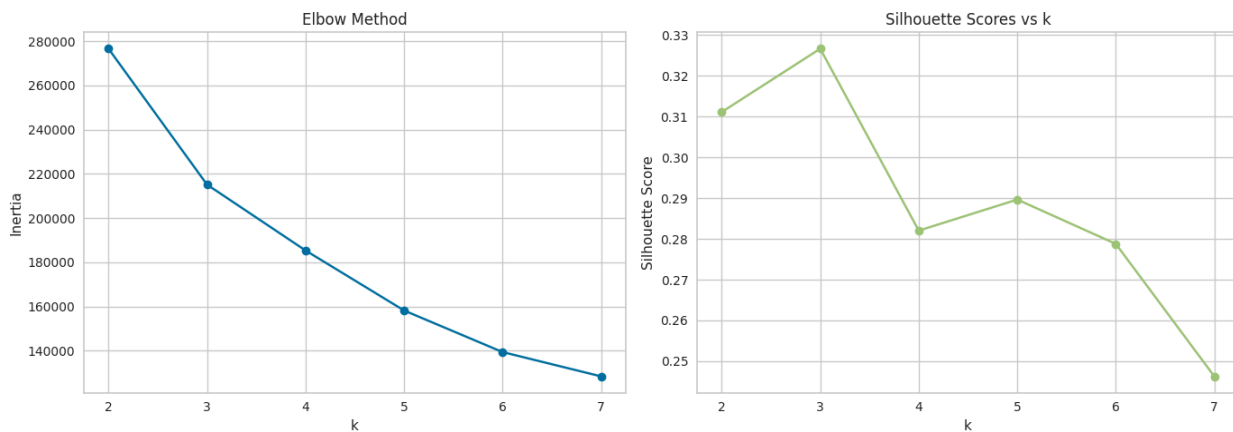
5. Clustering with Kmeans

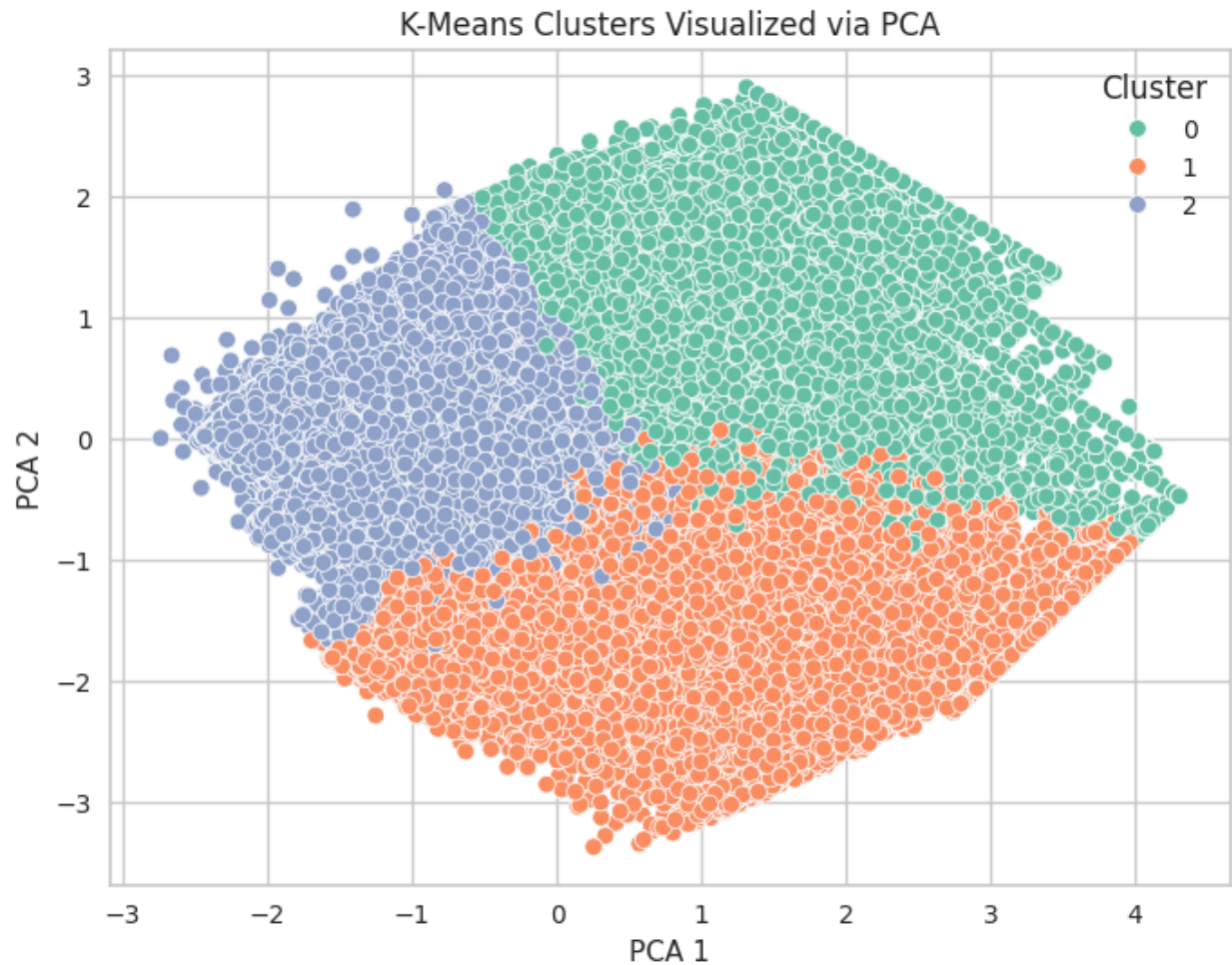
For clustering, I used K-Means on the numerical features. First, I standardized the data. Then, I used the elbow method and silhouette scores to determine the optimal number of clusters. The elbow suggested 4 clusters, and the silhouette score confirmed this. After applying PCA for visualization, I got a silhouette score of 0.227 and a Davies-Bouldin score of 1.513—not great.





To improve this, I dropped the frequency-encoded categorical columns and reran the process. This time, the optimal number of clusters was 3. The revised model yielded a silhouette score close to 0.33 and a Davies-Bouldin score of 1.28—an improvement. Cluster labels were added to the main dataframe.



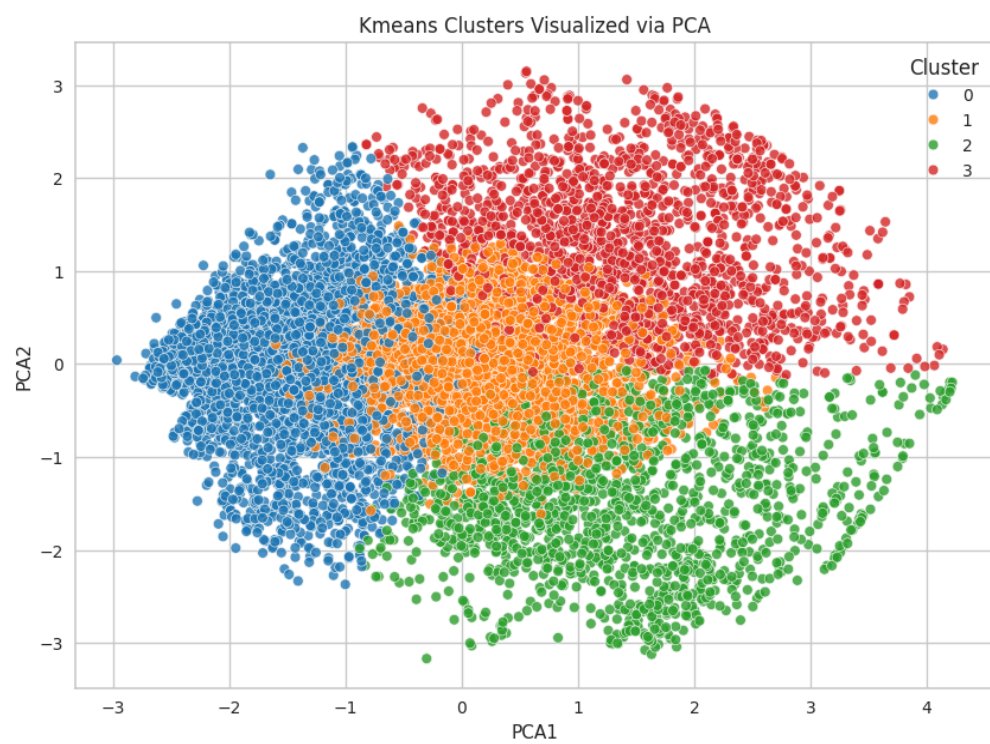
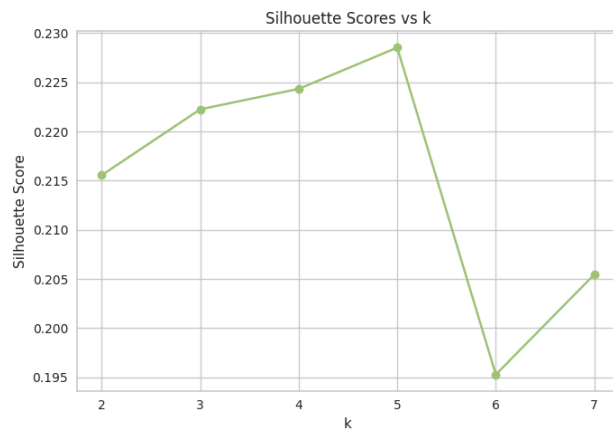
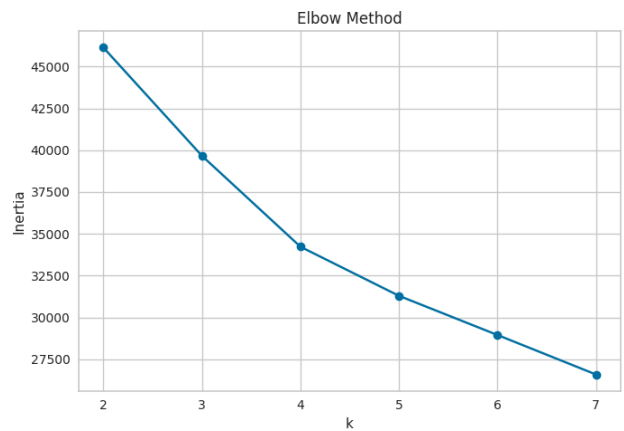


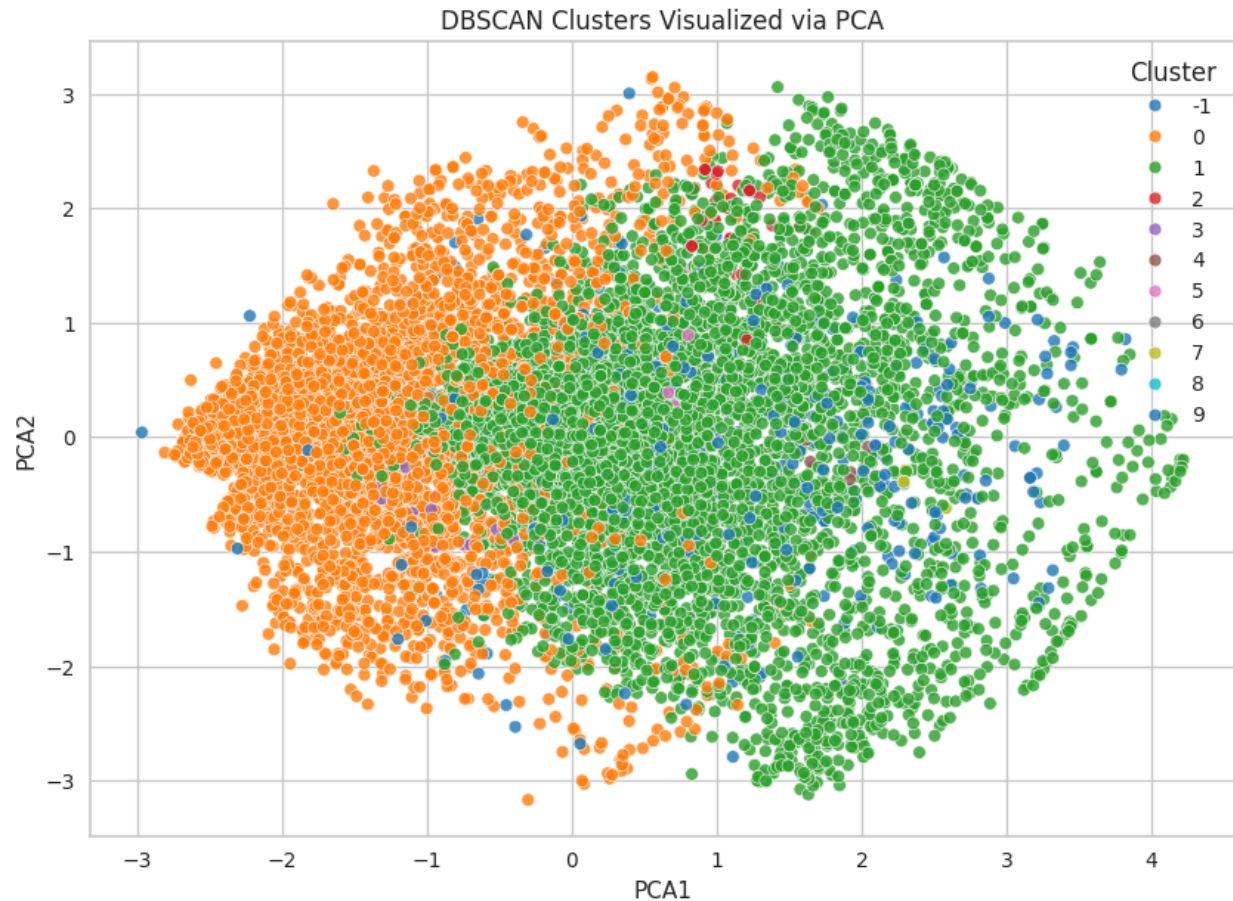
6. Model Variants

I wanted to try DBSCAN as a clustering alternative to K-Means. However, due to its computational cost, I ran it on a 10,000-row sample instead. To ensure fair comparison, I also applied K-Means on the same sample. The optimal cluster count for sample K-Means was 4.

The silhouette score for sample K-Means was 0.0224, with a DB score of 1.507. For DBSCAN, the silhouette score dropped to -0.33, but the DB score improved to 1.069, and it identified 9 clusters.

In terms of trade-offs, K-Means assumes spherical clusters and is sensitive to outliers. DBSCAN, on the other hand, handles arbitrarily shaped clusters and treats outliers as noise. However, DBSCAN's major downside is its scalability—it performs poorly on large datasets.





7. Cluster Profiling and Business Insights

After clustering, we performed a group-level aggregation by the cluster label to create a profile for each segment.

cluster	total_spe nt	n_orders	avg_revi ew_scor e	avg_freig ht_value	n_unique _categori es	avg_item s_per_or der	avg_ship ping_del ay	n_custo mers
0	224.270 012	1.15976 4	4.56619 5	26.0526 90	1.0	1.0	12.7945 11	23685
1	102.727 164	1.06875 2	2.90168 2	17.6383 77	1.0	1.0	15.91311 4	21352
2	76.2644 07	1.05025 6	4.77956 3	14.3414 13	1.0	1.0	8.21803 6	51019

- Cluster 0 consists of high-spending customers with an average total spend of R\$224. Despite having relatively long shipping delays (12.79 days) and high freight costs, their average review score is 4.57, indicating strong satisfaction. With over 23,000 customers in this segment, they appear to be loyal, possibly due to product quality or other value-added experiences. These users are ideal candidates for a VIP program that offers perks such as dedicated support or early access to sales. Faster shipping options could also enhance their satisfaction further.
- Cluster 1 includes customers with low average spend (R\$102) and the lowest review score (2.90), suggesting poor satisfaction. They also face the longest shipping delays (15.91 days). This group might represent first-time users who had a bad experience. Re-engagement strategies, such as apology discounts or promotional incentives, could help bring them back. Improving logistics transparency and delivery tracking would also go a long way in managing expectations and preventing dissatisfaction.
- Cluster 2 is the largest group, with over 51,000 customers. These users spend the least (R\$76 on average) but are the most satisfied, with an impressive average review score of 4.78. They benefit from fast shipping (8.22 days) and low freight costs. This segment is price-sensitive yet highly satisfied. To increase their lifetime value, the platform could experiment with upselling through free shipping thresholds or bundled offers. They could also be targeted with limited-time deals or flash promotions, leveraging their responsiveness and positive experience.